

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 What happen when laser interact with human body, define it with suitable diagram.
- Q.24 Write a note on therapeutic role of laser for Biomedical Engineering?
- Q.25 Explain applications of laser in Urology with its possible thermal, mechanical and photochemical effects.

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4th Sem / Medical Electronics

Subject : Laser Fiber Optics In Medicine

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1** First LASER was invented in:
- a) 1940 b) 1950
c) 1960 d) 1970
- Q.2** Which of the following is not a pumping process?
- a) Optical b) Electrical
c) Chemical d) Thermal
- Q.3** Which scientist first came up with the idea of stimulated emission:
- a) Alexander b) Issac Newton
c) Arthur Schalow d) Albert Einstein

Q.4 The Excimer laser produces light with what wavelength:

- a) UV b) Infrared
- c) Visible d) All of these

Q.5 The National ignition facility will use what type of laser for fusion power experimentation:

- a) Neodymium-glass b) Argon gas
- c) Rhodamine dye d) None of these

Q.6 Chemical laser use _____ to produce their beams:

- a) Excessive amount of electrical power
- b) Less amount of electrical power
- c) No electrical power
- d) All of these

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Range of cladding is μm is _____.

Q.8 O.F. works on principle of _____.

Q.9 Define Endoscopy.

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Q.10 Write name of any two materials used to make O.F.?

Q.11 Write down one application of O.F. in medical electronics?

Q.12 Define Absorption.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 What is O.F.?

Q.14 What is a laser diode?

Q.15 What is LED laser?

Q.16 Explain TIR with its suitable diagram.

Q.17 Write down 5(five) mechanical properties of O.F.?

Q.18 Define various standards required for OF.

Q.19 Write a note on laser used for skin related problems?

Q.20 Which laser is suitable to remove kidney stones?

Q.21 Write any five pros and cons of laser?

Q.22 What are the potential requirements for better laser according to its multidisciplinary applications?

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